Thoran Mani Sangeeth Patapalla

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Professional Summary

Final-year B.Tech student (AI & Data Science, 2027) building end-to-end Machine Learning and Generative AI systems – RAG pipelines, multi-agent LLM applications, and ML pipelines. Proficient in Python, Scikit-Learn, XGBoost, LangChain, CrewAI, and ChromaDB. Seeking full-time or internship roles as an **AI/ML Engineer or Generative AI Engineer**.

Education

B.Tech in Artificial Intelligence and Data Science — **2023 – 2027**
Sagi Rama Krishnam Raju Engineering College, Bhimavaram — *Current CGPA: 8.69/10*

Intermediate (MPC) — **April 2023**
SASI New Gen Junior College, Velivennu — *97.7%*

SSC — **June 2021**
Aditya School, Palakollu — *96.3%*

Technical Skills

Languages: Python, SQL
Machine Learning: Scikit-Learn, XGBoost, Feature Engineering
Generative AI / LLMs: LangChain, CrewAI, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, Hugging Face
Databases & Vector Stores: MySQL, ChromaDB
Tools & Platforms: Git, GitHub, Streamlit, Jupyter Notebook, Visual Studio Code
Core CS: OOP, DBMS

Projects

DocuChat – RAG Assistant | *Python, Streamlit, LangChain, ChromaDB, Hugging Face* | (GitHub Link)

- Engineered a full-stack **Retrieval-Augmented Generation (RAG)** application in Streamlit enabling natural-language question answering across **PDF, DOCX, TXT, Markdown, and CSV** documents.
- Designed the RAG pipeline using **LangChain, Hugging Face embeddings, and RecursiveCharacterTextSplitter**, with persistent vector storage in **ChromaDB** and automated document re-processing.
- Implemented top-8 semantic chunk retrieval integrated with **Google Gemini** for context-aware, multi-turn answers with maintained chat history across multiple documents.

Research Crew AI – Multi-Agent Research Assistant | *Python, CrewAI, Gemini, Serper, Streamlit* | (GitHub Link)

- Architected a **multi-agent Generative AI system** using CrewAI and Google Gemini, orchestrating specialized agents for web research, quality evaluation, data analysis, and report generation.
- Built an **iterative research quality-control loop** (up to 3 refinement cycles) scoring output on source quality, completeness, citation coverage, recency, and analytical value.
- Developed a **Streamlit interface** converting research topics into structured findings, quality reports, and analytical Markdown reports via **Serper**-powered web search.

Experience

Machine Learning Intern – APSSDC (Remote) — **May 2026 – Jul 2026**
Project – Predictive Maintenance using Python

- Built an end-to-end predictive maintenance pipeline using **Python, Scikit-Learn, and XGBoost** to predict equipment failure and classify failure types from industrial sensor data.
- Engineered features and optimized the ML pipeline with **ColumnTransformer, Pipeline, and GridSearchCV**, achieving a **0.83 F1-score** and **98.47% ROC-AUC** on a highly imbalanced dataset.
- Deployed the trained model as an interactive **Streamlit** web app for real-time failure-prediction demos. **Links:** GitHub | Live Demo

Certifications

- **Python Essentials Level 1 & Level 2 (2024):** Cisco Networking Academy.
- **Machine Learning using Python (2025):** Talent Trek.
- **HACKOVERFLOW Hackathon Participation (2024):** SRKR Engineering College